

Name: _____

Date: _____

Solve each problem. Show your work in the box.

Write your final answer on the answer line (or in the blanks shown).

Use the strategies you learned in Lesson 5.11.05.

1 Name the Quadrilateral

A quadrilateral has four right angles and all four sides equal 6 cm.

What is the name of this quadrilateral?

Square. _____

2 Properties of a Rectangle

A rectangle has a length of 10 cm and a width of 4 cm.

List three properties of a rectangle and find the perimeter.

Properties: 4 right angles; opposite sides are equal and parallel; diagonals bisect each other. Perimeter = $2(10 + 4) = 28$ cm.

3 Parallelogram vs. Rhombus

Shape A has opposite sides parallel and equal but angles that are NOT right angles. Shape B has all four sides equal and opposite sides parallel but no right angles.

Identify each shape and state one difference between them.

Shape A = parallelogram; Shape B = rhombus. A rhombus has all four sides equal; a parallelogram only requires opposite sides equal.

4 Trapezoid

A quadrilateral has exactly one pair of parallel sides. The parallel sides measure 12 cm and 7 cm. The height is 5 cm.

Name the shape and find its area. (Area = half $\times$ (sum of parallel sides) $\times$ height.)

Trapezoid. Area = $(1/2) \times (12 + 7) \times 5 = (1/2) \times 19 \times 5 = 47.5 \text{ cm}^2$. _

5 Reasoning: Sophia's Sorting

Sophia says every square is a rectangle, but not every rectangle is a square.

Is Sophia correct? Use the definitions of square and rectangle to explain.

Yes. A rectangle has 4 right angles and opposite sides equal. A square also has 4 right angles AND all sides equal, so it mee